Skill-Based Expert Routing for Multi-Label Harmful Language Detection

by

Mudit Baid

(Under the Direction of Dr. Ismailcem Budak Arpinar)

Abstract

The thesis is looking at the possibility of enhancing harmful-language moderation by coordinating multiple task-specific experts based on a symbolic routing framework that is interpretable. Instead of considering hate speech, offensive language, cyberbullying, and threats as one unified task, the proposed approach frames them as not independent, yet similar phenomena, and each is addressed by a special expert. To any input post, the system initially deduces a set of canonical harmful-language skills, and then employs these symbolic cues with the support of empirically-derived expert profile statistics to rank and pick experts to make the prediction. The framework is tested in scenarios of fractionated supervision, meaning there is no single dataset with unified multilabel annotations on all categories of targets. The experimental findings reveal that the task-specific experts perform well in their specific fields, and that the symbolic layer of skill reflects meaningful and interpretable harmful-language structure. Nevertheless, the symbolic space has considerable overlap among tasks and hence restricts strict expert separability and complex routing decisions. Compared to the baseline routing strategy, a top-k routing strategy based on Bernoulli can better rank experts and detect offensive language. However, there are still uneven end-to-end performance improvements across tasks, and routing behavior is not entirely selective. The main contribution of this thesis is a multi-expert framework of coordinated harmful-language detection that can be interpreted under heterogeneous supervision. Beyond showing its practical feasibility, the work presents a comprehensive description of its shortcomings, especially with respect to overlapping symbolic representations and limited routing selectivity.

INDEX WORDS: harmful language detection, multi-label classification, skill-based routing, expert models, social media NLP

Skill-Based Expert Routing for Multi-Label Harmful Language Detection

by

Mudit Baid

B.Tech., India

SRM Institute of Science and Technology, 2023

A Thesis Submitted to the Graduate Faculty of

The University of Georgia in Partial Fulfillment of the Requirements for the Degree.

Master of Science

Athens, Georgia

2026

Skill-Based Expert Routing for Multi-Label Harmful Language Detection

by

Mudit Baid

Major Professor: Ismailcem Budak Arpinar

Committee: Khaled Rasheed

Itai Himelboim

Electronic Version Approved:

Ron Walcott

Dean of the Graduate School

The University of Georgia

May 2026

Acknowledgments

I would like to express my sincere gratitude to my major professor, Dr. Ismailcem Budak Arpinar, for his constant guidance and steadfast support throughout this research. His feedback, encouragement, and clarity of direction were instrumental in shaping the ideas and execution of this thesis. I am also deeply thankful to Dr. Itai Himelboim for his valuable insights and mentoring, particularly in guiding my analysis and strengthening my understanding of harmful speech and social media research. I greatly appreciate his thoughtful questions and perspective, which helped refine both the framing and interpretation of this work. I would like to thank Dr. Khaled Rasheed for his technical insights, thorough review, and many helpful ideas. His careful comments and suggestions substantially improved the quality and rigor of the thesis. I gratefully acknowledge the Institute for Artificial Intelligence for providing the computational resources that supported the experiments in this research. Finally, I would like to thank Anusha, Madhulika, and all my fellow IAI graduate students for their support, and encouragement throughout this journey.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background and Motivation

Automatic identification of abusive language on social media has become a major research area in natural language processing. Online abuse can reduce healthy engagement, intensify social conflict, and create direct safety risks for individuals and communities. In response, researchers have developed a wide range of datasets and machine-learning models for abusive and malicious content detection. These resources often target related but distinct phenomena, including hate speech, offensive language, cyberbullying, and threats.

Several benchmark datasets have been especially influential in this area. For example, the Davidson et al. corpus distinguishes hate speech from offensive language, while OLID and OffensEval focus on offensive-language detection. Other corpora, such as those introduced by Vidgen et al., attempt to capture the evolving nature of online hate speech. Although these resources have advanced the field, they differ substantially in annotation guidelines, label definitions, sampling strategies, and class distributions. As a result, harmful-language data remains highly heterogeneous rather than uniform.

This heterogeneity creates a mismatch between training conditions and real-world moderation needs. In practice, abusive posts often combine multiple harmful behaviors. A single message may include profanity with targeted hostility, identity-based hate with threats, or interpersonal bullying expressed through offensive language. Despite this overlap, most benchmark datasets are designed for narrowly scoped tasks and expect a model to predict one type of harm at a time. They are rarely structured as reusable multidimensional multilabel taxonomies; instead, they are embedded in task-specific annotation schemes, even when related labels co-occur.

The challenge is not only that harmful-language categories overlap, but also that they are difficult to define consistently across datasets. Hate speech, offensive language, cyberbullying, and threats are related but not interchangeable. Annotation decisions often depend on how a dataset operationalizes protected targets, interpersonal aggression, contextual hostility, or intent to cause harm. Cross-dataset studies repeatedly show that models trained on one abusive-language dataset often generalize poorly to others. Differences in annotation criteria, topical distributions, and task design can significantly affect model performance. Therefore, simply combining datasets does not guarantee a coherent training signal for a single detection model.

The problem is further complicated by the limited availability of unified multilabel datasets. Some newer resources move in this direction. For instance, ETHOS includes hate-related multilabel annotations, and TweetEval consolidates several Twitter classification tasks under a shared benchmark. Even so, these datasets do not fully solve the broader problem of jointly modeling hate, offense, bullying, and threats under one comprehensive label system. This motivates a key research question: how can multilabel harmful-language detection be achieved when supervision is distributed across multiple task-specific benchmarks?

This thesis addresses that problem from a modeling perspective. Rather than assuming progress depends on first building a single large, continuously annotated multilabel dataset, this work examines whether separately trained task-specific models can be coordinated to produce consistent multilabel behavior. Specifically, it proposes a symbolic mixture-of-experts framework that integrates specialized detectors through an interpretable routing mechanism. In this formulation, the central challenge is not only identifying harmful content, but organizing heterogeneous supervision effectively and transparently.

1.2 Research Objective

The primary objective of this thesis is to evaluate whether meaningful multilabel harmful-language detection can be recovered from separately trained binary specialists when supervision is fragmented across heterogeneous benchmark datasets. More specifically, the thesis designs and evaluates a modular framework that combines task-specific experts for hate speech, offensive language, cyberbullying, and threat detection into a unified system that produces multi-expert judgments at the post level.

To achieve this goal, the proposed approach uses a symbolic mixture-of-experts architecture in which each harmful-language category is handled by a specialist model. Instead of relying on a single monolithic classifier, the framework coordinates multiple experts through an interpretable routing layer that determines which specialists should analyze each post.

A key feature of the framework is the use of explicit symbolic skills for expert selection. Rather than a purely latent gating mechanism, the system first predicts a set of human-readable skills that characterize the input text. These skills may include cues such as identity targeting, threatening language, general insult, or bullying-related behavior. The predicted skills are then combined with empirically estimated expert profiles that represent which skills each expert handles most reliably.

Using this information, the routing layer computes expert scores and activates the most relevant specialists for each post. Activated experts produce their own predictions, which are retained as multi-expert outputs rather than forced into a single consolidated label. In this way, multilabel behavior emerges from coordinated specialist interaction across heterogeneous training sources.

Conceptually, this design shifts away from the assumption that all harmful-language categories must be learned jointly within one model. Instead, each category is treated as a specialized detection task, and multilabel behavior is achieved through symbolic routing and profile-based coordination. The broader aim is to test whether this strategy provides a practical and explainable solution for harmful-language detection under fragmented supervision.

1.3 Research Questions

This thesis is guided by the following research questions:

RQ1. Can useful multilabel harmful-language detection behavior be recovered from separately finetuned binary experts when supervision is fragmented across heterogeneous benchmark datasets?

RQ2. Does skill-based routing improve expert selection compared with static or uniform expert invocation strategies?

RQ3. To what extent do explicit skills and profile-based routing improve interpretability in harmful-language detection relative to monolithic models?

RQ4. What limitations remain when fragmented benchmark datasets are used to approximate real-world multilabel harmful-language moderation?

These questions directly map to the core challenges addressed in this thesis. RQ1 evaluates feasibility: whether coordinated specialists can recover useful multilabel behavior. RQ2 examines whether symbolic routing provides measurable benefits over simpler coordination strategies. RQ3 focuses on interpretability, which is especially important in moderation settings where model decisions may need to be inspected and justified. RQ4 addresses practical limits, recognizing that fragmented datasets may remain biased, incomplete, or inconsistent despite improved modeling.

1.4 Organization of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 reviews prior research on harmful-language detection, abusive-language datasets, multilabel classification methods, ensemble strategies, mixture-of-experts architectures, and interpretable routing mechanisms. Chapter 3 presents the computational environment and theoretical background for the proposed framework, including large language models, parameter-efficient fine-tuning, and expert-based modeling. Chapter 4 details the methodology, including dataset preparation, expert construction, symbolic skill inference, expert profiling, routing strategy, and evaluation protocol. Chapter 5 reports experimental results and analyzes routing behavior, multilabel detection performance, and system limitations. Chapter 6 concludes with the main findings, implications for future harmful-language research, and directions for subsequent work.

Chapter 2

Literature Review

2.1 Heterogeneity of Harmful Language

Harmful language detection is often discussed as though it were a single classification problem, but the literature shows that it is more accurately understood as a family of related tasks whose boundaries only partially overlap. The evidence of foundational work in differentiating between hate speech and offensive speech is based on the idea that profane or insulting messages are not identity-directed hate (Davidson et al., 2017). The evidence on offensive language makes the situation more complicated with a hierarchical model of offense where the post is either offensive or the type and target of an offense is divided (Zampieri et al., 2019). Simultaneously, even the wider surveys of the abusive-language literature state that the datasets in this domain vary significantly in definition, annotation philosophy, and collection strategy, which implies that the field does not work with a single shared label space (Vidgen and Derczynski, 2020).

This abstract heterogeneity is even exaggerated when the notion of hate speech, cyberbullying, threats, and abuse through toxicity is viewed as a single entity. Hate speech is usually framed as hostility toward a protected or socially salient group, whereas cyberbullying is more often interpersonal and target-specific, and threat detection is narrower still because it centers on violent intent. These types can co-exist in the same message, but they are not interchangeable, and their working meanings differ between datasets. HateXplain specifies this by annotating both class labels and target communities and rationales, thus placing harmful-language classification as a problem in itself involving explanation and perspective, as well as not merely assigning labels (Mathew et al., 2021). Similar measurement-based efforts also view hate as an organized object and not a simple binary term, which again supports the opinion that harmful-language detection is conceptually and annotationally influenced to begin with.

In model design, this is important since a single post can have multiple harmful properties simultaneously. Hostility based on identity may be in the form of profanity, personal attacks, and innuendos of violence. However, benchmark resources tend to consider such phenomena as distinct work or any label structure that puts resources in the foreground and harms in the background. What has ensued is not just the issue of class imbalance; it is a problem of semantic overlap and mismatch with the annotation. This renders harmful-language detection a particularly challenging context of end-to-end multilabel learning in which the accessible supervision comes in the form of numerous partially aligned datasets as opposed to a single coherent benchmark (Vidgen and Derczynski, 2020; Zampieri et al., 2019).

2.2 Dataset Fragmentation

This fragmentation is manifested in the benchmark landscape. Davidson et al. (2017) is a classic corpus in hate-speech detection to distinguish between hate and offense, and subsequent literature (e.g., Vidgen et al., 2021) reacted to dataset brittleness by building dynamically generated hate-speech data that is expected to be more challenging and more robust. In the case of offensive language, OLID and OffensEval proposed a popular layered annotation scheme, and an offensive-language detector was added to a larger Twitter benchmark setup by TweetEval (Barbieri et al., 2020; Zampieri et al., 2019). These materials have been of great use in promoting individual sub-tasks, although they were not structured to offer a single, unified explanation of harmful language across categories.

Other datasets are adapted to richer supervision, although they are not a complete solution to the larger multilabel moderation problem either. ETHOS is a true multi-label hate-speech dataset, but with a multi-label framework that works in hate speech and not in the enlarged environment that encompasses offense, bullying, and threats (Mollas et al., 2022). HateXplain enhances interpretability through the addition of rationales and targets, yet the fundamental label model is hate-focused with offensive and normal content (Mathew et al., 2021). To be more precise, the motivation behind TweetEval is the fact that social-media evaluation is a disjointed set of heterogeneous tasks and protocols, which highlights the lack of a single standard benchmark even in the realm of Twitter-centered NLP research (Barbieri et al., 2020). These datasets collectively indicate that the field has numerous useful resources that are specialist in nature, but lacks a multilabel dataset that would be agreed upon by both communities and would encompass all major harmful-language classes in a well-balanced and conceptually coherent manner.

This conclusion is supported by cross-dataset evidence. Fortuna et al. (2021) demonstrated that hate speech, toxicity, offensive language, and abusive language models generalize unevenly over corpora, and some categories are more easily transferred than others. Arango et al. (2022) also reflected such a notion by arguing that seemingly high levels of hate-speech performance could be overstated by poor validation practices, and therefore good performance in-dataset did not necessarily imply effective conceptual learning.

Together, these studies indicate that current datasets cannot be viewed as parts of the same supervision space. Rather, they represent fragmented and heterogeneous supervision, which is precisely the condition that motivates this thesis.

2.3 Harmful Language Detection Models

The modeling literature has developed, to a great extent, in conformity with this benchmark structure. Previous research tended to use lexical or shallow supervised approaches, though the research rapidly transitioned to transformer-based fine-tuning as contextual encoders became the norm. One powerful such development is hateBERT: Caselli et al. (2021) have demonstrated that retraining BERT on abusive Reddit datasets produces higher gains than a general pretrained BERT on various abusive-language datasets. The bigger picture is that domain adaptation is helpful in specialist encoders since harmful-language cues are frequently subtle, community-specific, and lexically volatile. In this respect, powerful task-specific models have become a major accomplishment of the discipline.

At the same time, the success of specialist models does not eliminate the larger coordination problem. According to Vidgen et al. (2021), it is too easy to work with fixed datasets, and dynamic collection is needed to expose flaws that fixed benchmarks mask. Fortuna et al. (2021) also demonstrated that even powerful models do not generalize evenly when applied to new datasets due to discrepancies in underlying task formulations. Thus, the literature supports a dual conclusion: specialist models work well within their own task definitions, but harmful-language modeling remains fragmented because the data itself is fragmented. This opens the question of how separately trained detectors can be combined when a post displays overlapping harms across different benchmark traditions.

The current question has gained even more importance thanks to large language models. Zero-shot or minimally prompted LLMs are often able to detect explicitly harmful language, thus appearing to provide a universal solution to moderation. However, current studies suggest that prompt-only moderation is weak when the task requires awareness of target information, explanatory ability, or structured reasoning about harm. Roy et al. (2023) have found that LLM performance in hate-speech detection is significantly enhanced when target information and explanatory cues are available, which makes it clear that unstructured prompting is insufficient in subtle situations. This result supports an important suggestion for this thesis: despite high-performance foundation models, structured and interpretable intermediate representations remain necessary; rather than being replaced, they become more valuable.

2.4 From Ensembles to Expert Routing

As soon as harmful-language detection is theorized as a collection of overlapping, yet not identical tasks, the coordination problem arises as a major modeling challenge. Ensemble methods naturally address this by aggregating outputs of multiple models, often yielding greater robustness than single-model baselines. For harmful language, the reasoning is straightforward: different detectors can capture different linguistic, contextual, or target-related aspects of abuse, and combined predictions can improve overall performance. Normal ensembles do, however, generally treat coordination as an aggregation problem rather than an interpretable expert-selection problem. They disclose the final prediction, but rarely explain why one detector should receive more weight than others for a given post. This restriction is important in moderation, where diverse types of objectionable language may require different intervention schemes and where system behavior must be auditable.

Mixture-of-experts (MoE) architectures are more organized ways of coordinating specialists. More broadly, MoE systems in machine learning selectively activate a subset of experts in response to an individual instance, improving efficiency and specialization. However, traditional MoE routing is typically trained with latent neural gating, which masks the reasoning behind selecting particular experts. Such opacity is a weakness for harmful-language detection. If a post is better addressed by a hate-speech expert than an offense or threat expert, the system should explain why through recognizable textual properties, not only latent activations.

Specifically, the new Symbolic MoE framework applies especially closely to this topic because it switches routing to explicit skill-based coordination. Chen et al. (2025) support selecting experts based on symbolic competence according to the input and state that heterogeneous tasks usually require instance-dependent expertise instead of static task-level assignment. Even though their experiments focus on reasoning benchmarks rather than harmful-language moderation, the architectural principle transfers: routing can rely on interpretable intermediate signals rather than an opaque optimization layer. In moderation contexts, these signals can represent identity targeting, threatening language, interpersonal degradation, or profanity. Consequently, symbolic routing serves as a compelling bridge between specialist detectors and multilabel post-level behavior.

A significant technical reason why such architectures have become possible is parameter-efficient fine-tuning. Dettmers et al. (2023) showed that quantized low-rank adaptation allows large language models to be trained efficiently for specific tasks without full-model retraining. In specialist moderation systems, this matters because multiple task-specific adapters can share a common base model. In short, PEFT methods do not resolve the conceptual problem of harmful-language heterogeneity, but they make multi-expert solutions practical under realistic resource constraints. This practicality is vital to this thesis, because the proposed framework depends on multiple task-specific experts rather than forcing all supervision into one model.

2.5 Research Gap

All the literature points to an evident unresolved problem. Harmful-language research has generated strong task-specific datasets and detectors, but these resources remain scattered across platforms, annotation systems, label definitions, and constructs of interest. Existing multi-label datasets, though useful, are either limited to subdomains such as hate speech or integrated into platform-specific toxicity systems. Existing ensemble methods show that multi-detector coordination can improve performance, yet they do not fully solve how independently trained experts should be integrated into a comprehensible multilabel framework when supervision is heterogeneous. Cross-dataset studies consistently show this fragmentation is not superficial; it affects both validation and generalization (Arango et al., 2022; Fortuna et al., 2021; Vidgen and Derczynski, 2020).

That gap is filled by this thesis. Rather than assuming a single unified multilabel dataset, it investigates whether multilabel harmful-language behavior can be reconstructed from separately trained specialists. Its central premise is that fragmented supervision can be reframed as an architectural coordination problem rather than merely a data deficiency. By introducing symbolic skill inference as an intermediate layer between raw posts and expert activation, the proposed framework seeks to retain strengths of specialist models while making routing decisions explicit and interpretable. In this sense, the contribution extends beyond a conventional harmful-language classifier; it offers a symbolic, specialist-based framework for multilabel harmful-language detection under fragmented and heterogeneous supervision.

Chapter 3

Environment and Background Theory

3.1 Environment and Implementation Setup

This section details the computational environment and software infrastructure used to develop, fine-tune, and evaluate the language models in this study. Reporting these details supports transparency and reproducibility of the experimental setup.

3.1.1 Computing Hardware

All model training and experimentation were performed on a system with the following hardware:

- **GPU:** NVIDIA RTX A5000 with 24 GB VRAM
- **CPU:** used for data preprocessing and experiment control

The GPU provides sufficient memory and compute capacity to support parameter-efficient fine-tuning of large language models under quantized settings.

To support reproducibility and accessibility, all fine-tuned model checkpoints produced in this work are publicly available on the Hugging Face Model Hub:

https://huggingface.co/muditbaid/models

3.1.2 Software Environment

Experimentation was carried out with the following software stack:

- **Operating system:** Ubuntu 24.04.3 LTS
- **Python:** 3.10.19
- **Interactive workflow:** Jupyter notebooks
- **GPU acceleration:** CUDA-supported drivers
- **Deep learning framework:** PyTorch 2.9.0 (CUDA 12.8)

PyTorch was used to implement CUDA-enabled acceleration for tensor operations required during both training and inference of large language models.

3.1.3 Machine Learning Frameworks

Model development and training relied on several widely used LLM frameworks. The Hugging Face Transformers library served as the primary interface for loading and fine-tuning pre-trained language models. The PEFT library was used to implement parameter-efficient adapter-based fine-tuning.

The bitsandbytes library supported low-precision training and inference by enabling memory-efficient quantized model loading. LLaMA-Factory was used to simplify configuration and orchestration of LLaMA-based fine-tuning pipelines. Training procedures were executed with the TRL (Transformer Reinforcement Learning) library, specifically `SFTTrainer`, which provides utilities for supervised fine-tuning of large language models.

3.2 Background Theory

This section summarizes the theoretical concepts that motivate the modeling strategy in this thesis, including large language models, parameter-efficient fine-tuning, model quantization, and mixture-of-experts architectures.

3.2.1 Large Language Models

Large language models (LLMs) are neural models trained on large-scale text corpora to learn statistical language patterns. Most modern LLMs are based on Transformer architectures, which use self-attention to capture token-level dependencies within a sequence. The self-attention mechanism allows models to dynamically weight different tokens while forming contextual representations.

Instruction-tuned models extend pre-trained language models through additional training designed to improve instruction following in natural language. This extra stage improves model usability across a broad range of downstream tasks through prompt-based interaction.

The base model used in this study is LLaMA-3.1-8B-Instruct. The LLaMA family, developed by Meta, consists of decoder-only Transformer architectures for large-scale language modeling. The 8B model provides a practical balance between modeling capacity and computational efficiency for fine-tuning specialized classification tasks.

3.2.2 Parameter-Efficient Fine-Tuning

Full-parameter fine-tuning of large language models can require substantial computational resources and memory. Parameter-efficient fine-tuning (PEFT) addresses this constraint by introducing a small number of trainable parameters while keeping most pre-trained model weights fixed.

QLoRA (Quantized Low-Rank Adaptation) is a common PEFT approach that combines low-rank adaptation with quantized base models. Under this strategy, base model weights are frozen and stored in quantized form, while small trainable low-rank matrices are inserted into selected layers. This design significantly reduces trainable parameters and enables fine-tuning of large models on memory-constrained hardware.

3.2.3 Model Quantization

Model quantization reduces numerical precision used to represent neural network parameters. By encoding weights with lower-precision formats, quantization decreases storage requirements and can improve computational efficiency during training and inference.

In this work, BitsAndBytes is used to support 8-bit and 4-bit model loading. These lower-precision formats make it feasible to train and deploy large language models on GPUs with limited memory. QLoRA specifically relies on 4-bit quantization as a key component of its efficiency gains while preserving strong empirical performance.

3.2.4 Mixture-of-Experts Architectures

Mixture-of-experts (MoE) architectures consist of multiple specialized models (experts), where each expert learns different aspects of a task. Instead of sending every input through a single monolithic model, an MoE system uses a routing mechanism to select which expert or group of experts should process each input.

This design enables specialization at the expert level and supports better scalability on complex tasks. By selectively activating experts, MoE systems can represent broader linguistic patterns and achieve strong performance with lower effective computation than fully dense alternatives.

Chapter 4

Methodology

The identification of harmful language in the content of social media implies various phenomena that are closely tied and, at the same time, conceptually different, i.e., hate speech, offensive language, cyberbullying, and threats. Though the basic linguistic features of these tasks are similar, their annotation standards and class definitions, as well as data structures, do vary significantly. As a result, a single monolithic classifier that is trained on all the tasks might not effectively capture these differences. Following the symbolic mixture-of-exerts framework introduced by (Chen et al., 2025), this thesis adopts a skill-based expert routing architecture in which task-specific expert models are dynamically selected for each input instance.

4.1 Methodological Overview

The system operates through a three-stage pipeline. First, a symbolic skill-inference module extracts a set of interpretable conceptual signals from each social media post. Such signals are high-level linguistic or

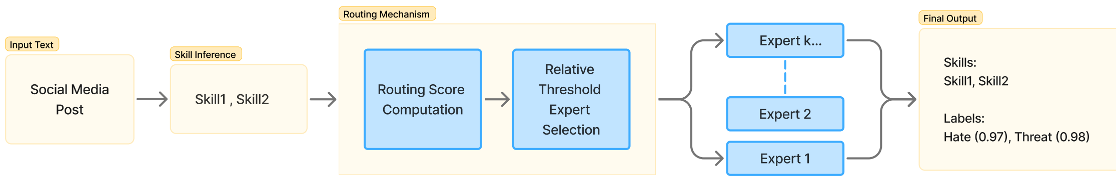


Figure 4.1: Skill Based Expert Routing Pipeline

behavioural cues, which are applicable to harmful-language detection. Second, an expert profiling stage evaluates the competence of each expert model under different skill conditions using labelled examples. Third, a routing mechanism combines the inferred skills with expert competence profiles to determine which expert models should be activated for prediction.

Rather than aggregating predictions into a single label, the system produces a multi-label output. For each input post, the routed experts independently generate predictions together with confidence scores. This design reflects the system's objective of multi-expert alerting: multiple specialised models may respond to the same input when overlapping harmful-language phenomena are present. The core methodological contribution of this work, therefore, lies in a routing mechanism that integrates interpretable skill extraction with dynamic expert selection.

4.2 Task Formulation

The aim of the proposed system is to detect harmful language in social media posts across several related categories. Where (x) refers to a social media posting in the form of raw text. The system maps each input post to output one or more task-specific labels produced by a set of expert models.

4.2.1 Input

The input to the system is a single social media post (x), represented as unstructured textual content.

4.2.2 Tasks

The methodology deals with 4 harmful-language detection tasks:

- Hate speech detection
- Offensive language detection
- Cyber-bullying detection
- Threat detection

Every task is addressed by an expert model that is finetuned on a specific dataset. In addition to the task-specific experts, the framework builds an expert profile for each expert. An expert profile represents how different experts perform on posts exhibiting different symbolic skill evidence, thereby providing skill scores rather than just a global accuracy value. For a new input social media post, the routing mechanism uses the skills provided by the skill-inference module and the expert profiles to compute a routing score for each expert. This routing score is used to select experts and divert the input post to those experts for task-specific prediction.

4.2.3 Output

The system produces predictions from one or more expert models. Every specialist comes up with a label in its respective task-dependent label space. Since the base model of the expert is a large language model, the confidence scores for each predicted label is calculated using the token logit-based scores. The skills inferred for the input post are also present in the output to make the predictions interpretale. Because multiple experts may be activated for a single post, the final output may contain several predictions with confidence scores corresponding to different harmful-language categories.

4.3 Datasets

The implemented system uses four datasets aligned with the four expert tasks:

- DynaHate, used for hate versus not-hate classification.
- TweetEval-Offensive, used for offensive versus not-offensive classification.
- Jigsaw Unintended Bias in Toxicity Classification, used for threat versus not-threat classification.
- SOSNet Cyberbullying, to be used in classifying bullying and not-bullying with subtype annotations.

Table 4.1: Datasets Summary

Dataset	Task family	Labels	Train count	Validation count	Test count
DynaHate	Hate	hate / not hate	32916	4100	4120
TweetEval-Offensive	Offense	offensive / not offensive	7882	918	480
SOSNet Cyberbullying	Bully	bully / not bully	35883	3986	N/A
Jigsaw Threat	Threat	threat / not threat	3500	1334	1334

Note. The SOSNet cyberbullying contains train and validation splits only.data/.

All datasets are converted into a unified instruction-style JSONL format containing the fields system, instruction, input, and output. The original post text is preserved unchanged as the input field.

Task-specific prompts are placed in the system and instruction fields, while the output field contains the normalised gold label for the task. For the first three datasets, the target outputs are binary textual labels (e.g., hate vs. not hate). The cyberbullying dataset employs a structured output format:

```
label:  bully|not_bully; type:  age|gender|ethnicity|religion|none
```

In this format, subtype information may be expressed with a uniform schema of labels. The Jigsaw dataset has the labeling schema: *very toxic*, *toxic*, *hard to say*, *not toxic* with toxic subtype attributes: *identity_attack*, *insult*, *obscene*, *sexual_explicit*, *threat*. This work takes a subset of the mentioned Jigsaw dataset where only posts with positive threat label are selected. To balance the threat subset, equal number of neutral posts are selected.

There exists a standard process in dataset preprocessing. First, the post text is retained as the input field. Secondly, task-specific system and instruction prompts are introduced. Third, the gold label is normalised to the task-specific output schema. Finally, the processed record is stored as an instruction-style JSONL entry.

4.3.1 Profiling, Validation and Test Data Pools

Table 4.2: Data Pool Composition

Pool name	Total	DynaHate	TweetEval	SOSNet	Jigsaw	Purpose
Profile pool	14000	3500	3500	3500	3500	Build expert profiles from inferred skill annotations.
Validation pool	3672	918	918	918	918	Calibrate the routing threshold parameter α.
Test pool	1920	480	480	480	480	Evaluate the final pipeline on held-out examples.

The suggested methodology will be based on three separate data pools, which will have different purposes in the entire pipeline. This separation is necessary in ensuring that routing decisions are not made based on the same information on which evaluation was made. First, a profile pool is used to estimate expert behavior under different symbolic skill conditions. Second, a validation pool is used for routing calibration, top-k selection and inference-time evaluation. Third, a test pool is reserved for held-out evaluation of the final routed system.

In the current experimental configuration, the profile pool contains 14,000 posts constructed by sampling 3,500 examples from the training split of each of the four task datasets: DynaHate, TweetEval-Offensive, SOSNet Cyberbullying, and Jigsaw Threat. The validation pool contains 3,672 posts (918 per dataset), while the test pool contains 1,920 posts (480 per dataset). This separation is methodologically important because expert profiles (Section 4.5) are computed exclusively from the profile pool and are fixed prior to validation and test inference. Consequently, the routing mechanism (Section 4.7) operates using the expert profiles that is independent of the data used for final evaluation, reducing the risk of information leakage.

4.4 Expert Model Construction

The individual expert models are deployed as Low-Rank Adaptation (LoRA) adapters, which are fine-tuned on top of a base large language model, Meta-Llama-3.1-8B-Instruct. The active expert set is made up of four adapters:

- Hate Expert, finetuned on the DynaHate dataset
- Offensive Expert, finetuned on TweetEval-Offensive dataset

- Bully Expert, finetuned on SOSNet Cyberbullying dataset
- Threat Expert, finetuned on the Jigsaw Threat dataset

Table 4.3: Expert Summary

Expert name	Dataset	Model	Adapter	Output label
Hate Expert	DynaHate	Meta-Llama-3.1-8B-Instruct	QLoRA	hate / not hate
Offense Expert	TweetEval Offensive	Meta-Llama-3.1-8B-Instruct	QLoRA	offensive / not offensive
Bully Expert	SOSNet Cyberbullying	Meta-Llama-3.1-8B-Instruct	QLoRA	bully / not bully
Threat Expert	Jigsaw Threat	Meta-Llama-3.1-8B-Instruct	QLoRA	threat / not threat

During inference, the base model architecture and task-specific adapters are loaded to the gpu when the corresponding expert is selected by the routing mechanism. Each expert produces predictions within its own label space, and the outputs are normalized to ensure consistent evaluation. Expert predictions are generated using greedy decoding, and label confidence scores are computed by evaluating the log-probability of candidate label strings defined for each task.

4.4.1 Fine-Tuning Procedure

All the 4 task specific expert models are fine-tuned using QLoRA efficient PEFT approach. The shared pre-trained model Meta-Llama-3.1-8B-Instruct, while each task is finetuned as a LoRA adapter. The LoRA parameters are the following: LoRA rank ($r = 8$) LoRA scaling factor ($\alpha = 16$) LoRA dropout ($= 0.05$) Adapters are applied to the 4-bit quantized base model, including query, key, value, output, and feed-forward projection layers. Finetuning employs 4-bit quantisation through the bitsandbytes library with the NF4 quantisation scheme and double quantisation. Computation is performed using the bfloat16 datatype on CUDA-enabled hardware. Gradient checkpointing is enabled to reduce memory

Table 4.4: Shared Training Hyperparameters

Parameter	Value
Base model	Meta-Llama-3.1-8B-Instruct
Fine-tuning method	QLoRA
LoRA rank	8
LoRA alpha	16
LoRA dropout	0.05
Quantization	4-bit bitsandbytes (bnb)
Training stage	Supervised fine-tuning (SFT)
Optimizer	paged_adamw_32bit
Learning-rate scheduler	cosine
Number of epochs	3
Per-device train batch size	4
Learning rate	2.0e-5
Maximum sequence length	2048
Warmup ratio	0.05
Gradient accumulation steps	8
Gradient checkpointing	enabled
Template	llama3
Precision	bf16

consumption during training. All expert models are finetuned for three epochs with instruction-style classification prompts rendered using the llama3 chat template. The learning rate scheduler is cosine, and optimisation is done using the Paged AdamW optimiser. All the experts have a maximum sequence length of 2048 tokens.

4.5 Expert Profiling

Expert profiling is conducted as a preparation stage whose purpose is to estimate how reliable each task-specific expert is under different skill evidence. Instead of making a direct classification of posts, this step builds up an empirical map between interpretable skills and expert performance. The resulting expert profiles provide the prior knowledge later used by the routing mechanism. The process of profile construction takes four steps. First, the profile pool described in Section 4.3 is assembled from the training splits of the four datasets. Second, symbolic skill-inference (Section 4.6) is applied to these posts so that each example is associated with 0 to 5 canonical skills. Third, every expert model is tested on the respective task subset of the profile pool. Lastly, the comparison of expert predictions and normalized gold labels is made and produced into per-skill statistics of correctness. The expert profiling module retains all the expert predictions on their respective task subsets and the canonical skills provided by the skill-inference module. The module now aggregates these results skill by skill, to assign a score to each expert, based on how frequently the expert encounters the skill and how often the expert predicts the task label correctly when that skill exists. Normalized profile score is then calculated as:

$$\text{score} = \frac{2 \times \text{correct} - \text{total}}{\text{total}}$$

This generates the values within the range of -1 to 1. The positive values mean that the expert will likely do well in the situation when the skill is available, whereas the negative values mean that the expert will not have a strong performance in that condition. Besides these skill scores, each expert profile also archives a global accuracy value that is calculated on the subset of the profile-pool of the expert. This is used to serve as a prior estimate of the reliability of experts and is subsequently used in the routing decision. The role of expert profiling in the overall methodology is therefore not to perform classification directly, but to provide a statistically grounded basis for expert selection. These are calculated only once and then re-utilized in validation as well as test-time routing.

4.6 Skill Inference

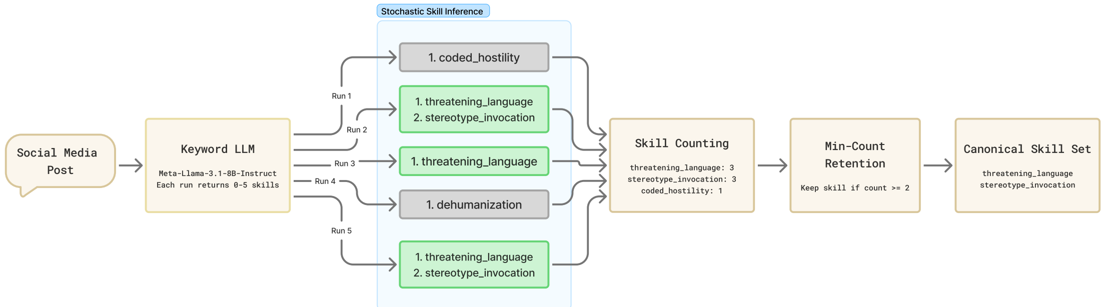


Figure 4.2: Skill Inference Workflow

Symbolic skill-inference module is the component that converts raw post text into an interpretable intermediate representation used by the expert profile module and routing mechanism. The system employs a fixed vocabulary of 11 canonical skills, including identity targeting, dehumanization, stereotype invocation, coded hostility, non-targeted profanity, general insult, threatening language, identity-based bullying, appearance-based bullying, age-based bullying, and sexual harassment. The skill-inference module entails prompting the keyword llm to present 0-5 skills from the canonical skills set solely based on evidence present in the input post. The keyword model used in the current pipeline is Meta-Llama-3.1-8B-Instruct, chosen for its strong instruction-following capability. To enhance reliability, inference of the skills is done by repeating stochastic sampling in place of one deterministic generation. Within this context, stochastic means sampling based decoding, in which repeated executions of the same input can obtain slightly different results. The model is sampled with a temperature of 0.2 and nucleus sampling (top-p) of 0.95 with 5 independent runs per post implemented. When generated, the predicted skills are generated and aggregated. Not all the skills are retained but only those that are present in two or more of the 5 runs. The consensus based methodology guarantees that there is less variation in single-runs and more consistent symbolic cues about downstream routing. The structured output:

```
Skills: ["skillone", "skilltwo"]
```

This process is applied in two phases of pipeline. During Expert Profiling, it is applied to the profile pool to annotate examples prior to expert profiling (Section 4.5). At the inference stage, it is applied to each incoming validation, test, or deployment post prior to routing (Section 4.7). The same symbolic vocabulary in both stages is necessary to make sure that there is consistency between profile construction and inference-time decision making.

4.7 Routing Mechanism

This stage is the most important step in the pipeline. A social media post is not just raw text, but has tagged symbolic skills (Section 4.6) and expert profiles (Section 4.5) linking experts to their skill strengths along with it. With this information, the router determines which expert models should be selected for said social media post connecting information retrieval to multilabel classification. Because the overall framework is designed for harmful-language detection under fragmented supervision, the routing mechanism is central to the methodology, and determines where the correct experts are activated for hate, bully, offense and threat detection. The routing mechanism has three responsibilities: compute routing scores for each expert using skill evidence and expert profiles, produces an expert ranking for each post, controls how many experts are actually called for inference. The proposed methodology experiments with 2 different routing systems. The first one acts as the baseline and follows the routing system of the Symbolic-MoE framework (Chen et al., 2025) which uses expert prior scores and skill log-odds. The second is a bernoulli router introduced to prioritize expert ranking by modelling both presence and absence of skills in the input post. Both the routers are explained in detail below.

4.7.1 Baseline Router

The routing mechanism of the baseline router is straightforward: use the skill-condition statistics from the expert profile to choose the experts for the input post. The routing decision can be explained as a combination of expert prior competence and support from the inferred skills.

Scoring Rule:

Each expert receives:

- A global prior-log-odds score derived from overall expert correctness.
- A set of per-skill-log-odds values derived from skill-conditioned statistics.

For expert *e* and skill set *S*, the routing score is:

$$\text{score}(e) = \text{prior}(e) + \sum_{s \in S} \text{logodds}(s, e)$$

where the skill-log-odds is calculated using Laplace smoothed profile counts:

$$\text{logodds}(s, e) = \log \left(\frac{P(\text{correct} \mid s, e)}{1 - P(\text{correct} \mid s, e)} \right)$$

The baseline router only uses skills that are inferred. Absent skills do not contribute to the baseline routing system.

Selection Rule

The experts with positive routing scores are treated as candidate experts. The router now applies a relative threshold rule: experts are selected if their routing score is atleast a fraction α of the maximum routing score for that post.

The hyperparameter α is calibrated on the validation pool to balance predictive performance and routing selectivity. The calibration measures overall post-level accuracy with average number of experts selected per post. The calibration results of α are shown in Table 4.5.

Table 4.5: Calibration of α on validation pool

α	Overall Accuracy	Avg. Experts/Post
0.1	0.8516	3.410
0.2	0.8516	3.410
0.3	0.8516	3.410
0.4	0.8516	3.086
0.5	0.8399	3.013
0.6	0.8358	3.000
0.7	0.8355	2.996
0.8	0.7775	2.801
0.9	0.7086	1.803

The average number of experts per post is close to 3 for most values of α which shows that the baseline router is permissive but also shows that different task-specific expert have similar routing scores which makes it difficult to route selectively. The value for α is set to 4 where the overall accuracy is maximum with least average experts per post for the same accuracy score. These calibration results promote evaluation of routing behaviour and also scrutiny of domain skill evidence in social media posts. For example, a sample post from DynaHate dataset could show evidence of gender-based bullying and threatening speech but the gold label would still be hate speech. During inference, the conceptual skill increases the possibility o triggering bully expert and threat expert along with hate expert. This example doesnt imply a routing mechanism failure but instead highlights that fragmented task-specific datasets have overlapping skills.

In the case where no skill is inferred for a post, the baseline relies on piror-only scores. Since, several experts have positive prior scores, the no-skill condition often leads to broad routing. This also increases the average experts per post.

The baseline router exhibits interpretability. It's decisions are easy to explain because each routed expert is supported by a visible combination of expert prior and observed skills' strength. On the other hand, the downside of this router is clustered routing scores and broad selection as it only accounts for the skills present in the post.

4.7.2 Bernoulli Router

The bernoulli router was introduced to improve selective routing by taking the full skill-pattern into consideration. Instead of scoring the experts only based on what skills are inferred, it compares the binary skill pattern of the post against the learned skill distribution of each expert. This utilizes information from the skills that are absent in the post aswell.

Bernoulli Router Training:

The bernoulli router is trained on the profile pool after skill inference which includes the post, its inferred skills, and the task label of the post. During training, for each expert *e* and skill *s*, the router estimates the probability that each skill appears in posts associated with that expert's task dataset.

Let:

- N_e: be the number of training rows associated with expert e.

- $C_{e,s}$: be the number of those rows where skill s is present.

Then the Bernoulli parameter for skill s under expert e is estimated with Laplace smoothing as:

$$P(s|e) = \frac{C_{e,s} + 1}{N_e + 2}$$

The expert prior is also estimated with smoothing:

$$P(e) = \frac{N_e + 1}{N + |E|}$$

where N is the total number of router-training rows and $|E|$ is the number of experts. This produces an expert-specific Bernoulli profile over the full symbolic vocabulary.

Scoring Rule:

At inference time, the inferred skills are converted into a binary vector over the skill vocabulary, and each expert is scored according to how well that skill pattern matches the expert-specific Bernoulli profile. If a skill is present, it contributes $\log P(s \mid e)$. If a skill is absent, it contributes $\log(1 - P(s \mid e))$. The full score is:

$$\text{score}(e) = \log P(e) + \sum_{s \in S_{\text{present}}} \log P(s \mid e) + \sum_{s \in S_{\text{absent}}} \log(1 - P(s \mid e))$$

textbfSelection Rule: Unlike the baseline router, the bernoulli router acts as a ranking model and thus is used with top-k selection. The parameter k is also calibrated on validation pool prioritizing routing precision instead of overall accuracy. Top-2 criteria is selected because it improves gold-expert recall significantly while preserving routing top-1 accuracy.

In no-skill evidence cases, the bernoulli router does not fallback to expert priors, but instead, still ranks each expert based on the absence pattern over the complete skill set S. This enables bernoulli to produce ranking even under sparse symbolic evidence.

$$\text{score}(e) = \log P(e) + \sum_{s} \log(1 - P(s \mid e))$$

Bernoulli router exhibits better ranking discrimination as a result of using both presence and absence of skills. Although, this router is comparatively less intuitive than the baseline router.

4.8 End-to-End Pipeline

At runtime, a raw social media post is first converted into a symbolic skill representation. These inferred skills are then passed to the routing module, which combines them with the precomputed expert profiles obtained during the offline profiling stage in order to select one or more experts for execution. The chosen professionals analyze the original posting in their specific task label spaces and provide forecasts with the graduated confidence scores. Because the framework is designed for multilabel harmful-language behaviour under fragmented supervision, the outputs of the activated experts are retained as parallel task-specific predictions rather than merged into a single aggregated label. The final result, therefore, consists of the inferred skills and the corresponding expert predictions with confidence values.

4.9 Evaluation Protocol

Evaluation is done on both validation and test pools. Since the routing mechanism can engage many experts on a given post, the definition of correctness should be such that it takes into consideration the multi-expert predictions. Evaluation is used in this work and it takes into account the predictions of routed experts. In the case of each post, the experts are ranked based on their routing scores. A prediction is considered correct if the gold normalized label is produced by any of the routed experts. This assessment plan resembles the multi-expert design of the system. Practically, multiple specialists can be the ones to react positively to the same input and evaluation will determine whether the appropriate specialist model would be found among the most relevant models. Performance is reported using overall accuracy routing precision metrics and F1 scores for each task.

Chapter 5

Results and Discussion

This chapter reviews the suggested framework at three levels that are complementary to each other. First, it considers the quality of the task-specific expert models per se, as the process of routed performance does not make sense unless the underlying experts are also reliable. Second, it examines the symbolic layer and the profile-based evidence which links posts to the expert selection. Third, it uses the end-to-end routed system on held-out validation and test pools, both routing-oriented behavior and post-level predictive performance.

5.1 Evaluation Scope

The evaluation scope must be taken with discretion. Since the framework is built based on four benchmark tasks having distinct sets of labels, the current study is not a fully supervised multilabel assessment in a strict sense of one shared set of annotated gold data. Rather, it evaluates routed multi-expert harmful-language detection under fragmented supervision. The central question is therefore not whether a single model has solved unified multilabel moderation, but whether symbolic skill inference and expert routing can meaningfully coordinate strong task-specific experts across related harmful-language phenomena.

The findings are ambivalent yet informative. The expert models are strong within their own domains, and the symbolic layer captures interpretable harmful-language structure across datasets. At the same time, routing remains only partially selective, reflecting substantial overlap in the symbolic space and incomplete expert separability. The chapter thus highlights the strong point of the framework as well as the intrinsic domain-level overlap of skills that constrains routing selectivity.

5.2 Task-Specific Expert Performance

To determine the routed framework as competent in its entirety, there is a need to determine that the underlying expert models are competent enough in their respective domains of work. The offered architecture implies that hate, offensive language, cyberbullying and threats each can be presented by the trusted experts and that the key challenge at the system level is the selection of the experts and not the learning of each task separately. The benchmark comparison of each expert with its task-specific baseline is presented in Tables 5.1 through 5.4.

Table 5.1: Comparison of Hate Expert against DynaHate Benchmarks

Model	Macro-F1	Setting
Hate Expert	0.914	Fine-tuned
RoBERTa (R4 Target; Vidgen et al., 2021)	0.759	Supervised
GPT-4o (Kim et al., 2025)	0.832	Zero-shot
GPT-4o-mini (Kim et al., 2025)	0.835	Zero-shot
GPT-5.1	0.810	Zero-shot

Table 5.2: Comparison of Bully Expert against SOSNet Benchmarks

Model	Accuracy	Macro-F1	Setting
Bully Expert	0.951	0.950	Fine-tuned
SBERT + SOSNet (GCN; Wang et al., 2020)	0.927	0.926	Supervised
SBERT + MLP (Wang et al., 2020)	0.915	0.915	Supervised
DistilBERT + SVM (Wang et al., 2020)	0.919	0.919	Supervised

Table 5.3: Comparison of Offense Expert against TweetEval-Offensive Benchmarks

Model	Macro-F1	Setting
Offense Expert	0.828	Fine-tuned
TimeLM-21 (RoBERTa, Twitter-retrained; Loureiro et al., 2022)	0.822	Fine-tuned
RoBERTa-Retrained (Twitter; Barbieri et al., 2020)	0.816	Fine-tuned
Ensemble (CNN + BLSTM + BGRU; Zampieri et al., 2019)	0.807	Supervised
fastText (Barbieri et al., 2020)	0.734	Supervised
BLSTM (Barbieri et al., 2020)	0.717	Supervised
SVM (Barbieri et al., 2020)	0.523	Supervised

Table 5.4: Comparison of Threat Expert against Jigsaw Threat Benchmarks

Model	Accuracy	Setting
Threat Expert	0.964	Fine-tuned
BERT (Khan, 2023)	0.890	Fine-tuned
Ensemble (RF + Bagging + GB + AdaBoost; Islam et al., 2023)	0.898	Supervised
Logistic Regression + TF-IDF (Khan, 2023)	0.571	Supervised

The expert models do well in all the four tasks. The hate expert is competitive in regards to the known hate-speech thresholds, the offense expert is in the strong range of the TweetEval-Offensive benchmark models, the bully and threat experts work the best on their respective benchmarks as shown in tables 5.1-5.4. These benchmark results show that the experts are not weak auxiliary components included merely for architectural convenience; rather, they are credible task-specific detectors trained on standard harmful-language benchmarks.

This conclusion is supported by the profile-pool statistics. The hate expert has an accuracy of 0.9540, the offense expert 0.8029, the bully expert 0.9634, and the threat expert 0.9729 on the 3,500 examples each profile subset that is used in each of the tasks. The three of the four experts are thus very reliable in task-congruent held-out data whereas the offense expert is relatively weak though still significantly higher than trivial performance. The significance of this rank in the strength of an expert is that it affects the development of the profile and the future guided behavior.

All these findings contribute to the main assumption of the framework. The fact that the experts are already powerful on the island means that any subsequent failures of the routed system can not be laid to the weakness of learning tasks as their main cause. The fact that the expert in offense outperforms the offense task benchmarks but not routing also indicates that part of the increase in the system level of difficulty is due to actual task difficulty of offensive-language detection and not routing.

5.3 Skill Distributions Across Datasets

The interpretable medium of representation of the proposed framework is the symbolic skill layer. Rather than routing directly from raw text, the system first infers a small set of canonical harmful-language skills, which are then used to characterize posts and support expert selection. Analyzing how these inferred skills are distributed across datasets is therefore important for determining whether the symbolic layer captures meaningful harmful-language structure. The general distribution of the skills in all the four datasets is represented in Figure 5.1, whereas the complete per-dataset distributions are presented together in Figure 5.2.

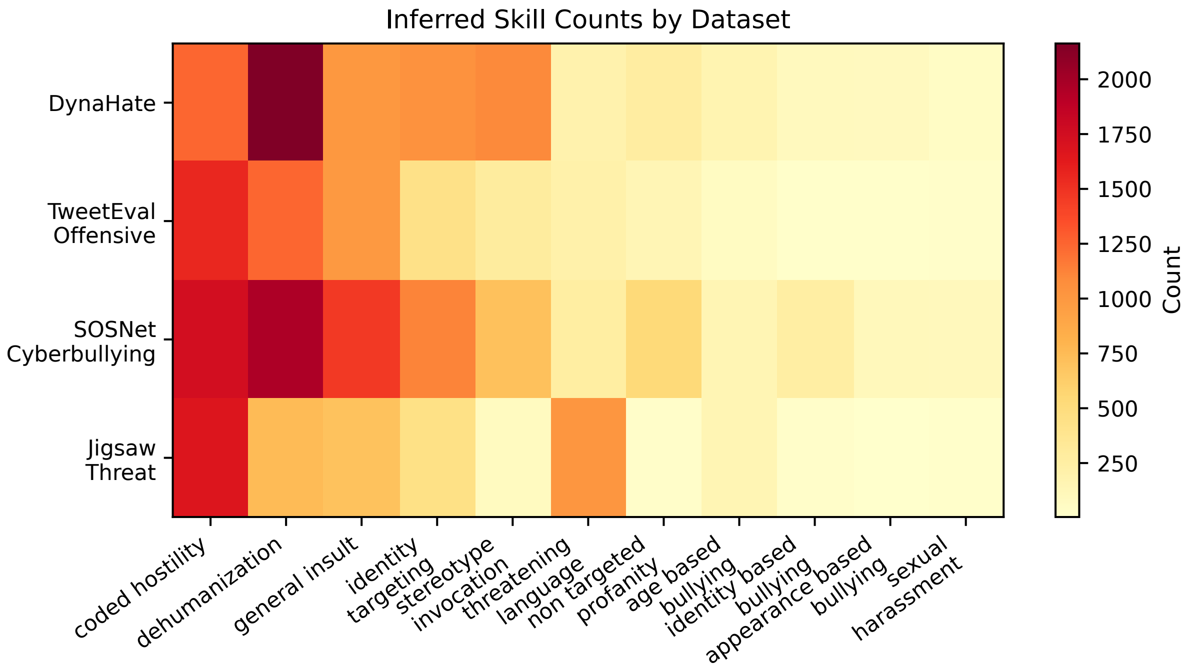
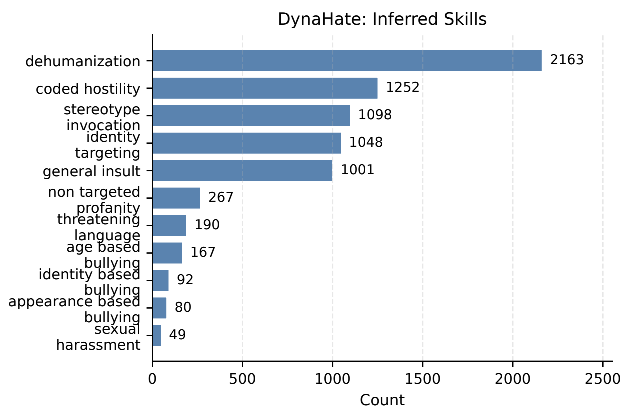
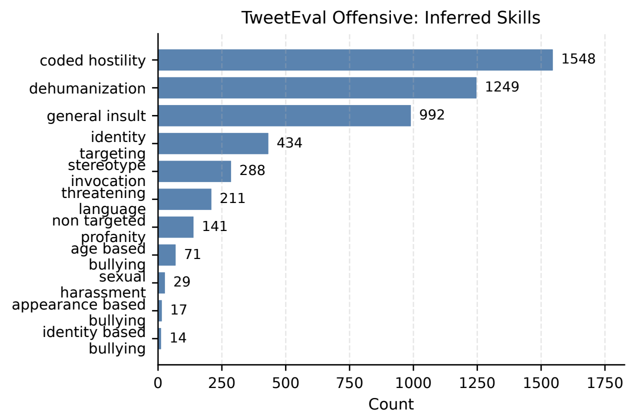


Figure 5.1: Dataset Skill Heatmap

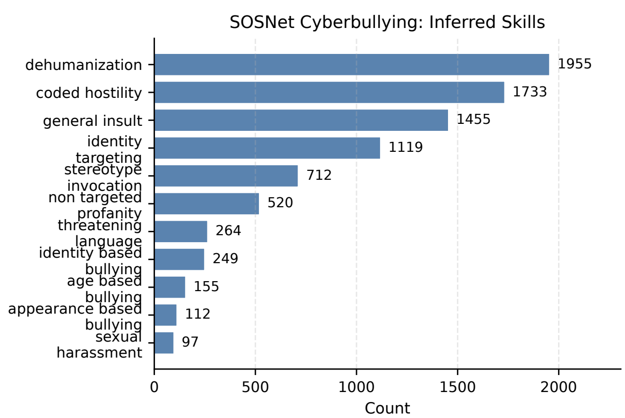
The derived skills are not shared randomly. The DynaHate dataset is dominated by patterns such as dehumanization, coded hostility, stereotype invocation, identity targeting, and general insult, which is consistent with the semantics of hate-speech detection. The Jigsaw Threat dataset places greater emphasis on threatening language together with coded hostility, reflecting its narrower focus on explicit threat-related expression. The SOSNet Cyberbullying and TweetEval-Offensive datasets also show recognizable



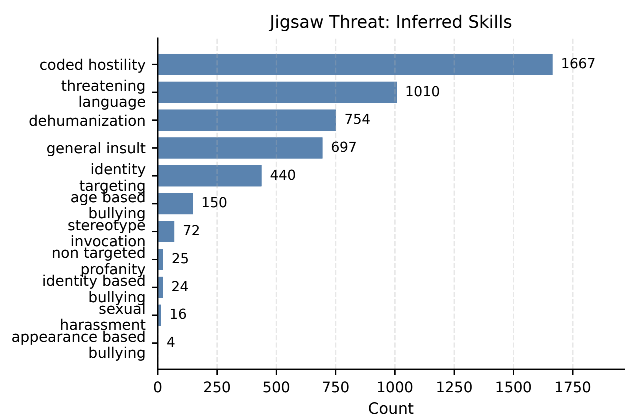
(a) DynaHate Skill Distribution



(b) TweetEval Offensive Skill Distribution



(c) SOSNet Cyberbullying Skill Distribution



(d) Jigsaw Threat Skill Distribution

Figure 5.2: Skill distributions across datasets.

task-related patterns, although they rely more heavily on broad abusive signals such as general insult, coded hostility, and dehumanization.

Meanwhile, there is a high degree of overlap in the distributions. Several of the most frequent skills, especially coded hostility, dehumanization, and general insult, occur at high rates across multiple datasets rather than serving as task-exclusive markers. This indicates that the symbolic layer captures a shared harmful-language substrate more clearly than it captures cleanly separated task boundaries. Other skills are represented sparsely, but the broad, reusable cues which tend to take up much of the symbolic space can realistically trigger more than one expert.

There are direct routing implications of this overlap. The symbolic layer is meaningful because it provides an interpretable and task-relevant representation of harmful-language patterns, but the skill space is only partially separable. As a result, symbolic evidence can guide expert ranking without guaranteeing sharply selective expert discrimination. The main worth of the symbolic layer in this sense is organised interpretability and part-way routing signal as opposed to real decomposition of harmful language into mutually exclusive fields of expertise.

5.4 Expert Profile Characteristics

After symbolic skills are inferred, the framework constructs expert profiles that summarize how reliably each expert performs under different skills. These profiles are intended to provide the empirical link between the symbolic layer and routing behavior. Instead of defining experts by the global task accuracy, the profile stage documents the nature of expert correctness as affected by select symbolic skills. Figure 5.3 visualizes the aggregate profile structure across experts.

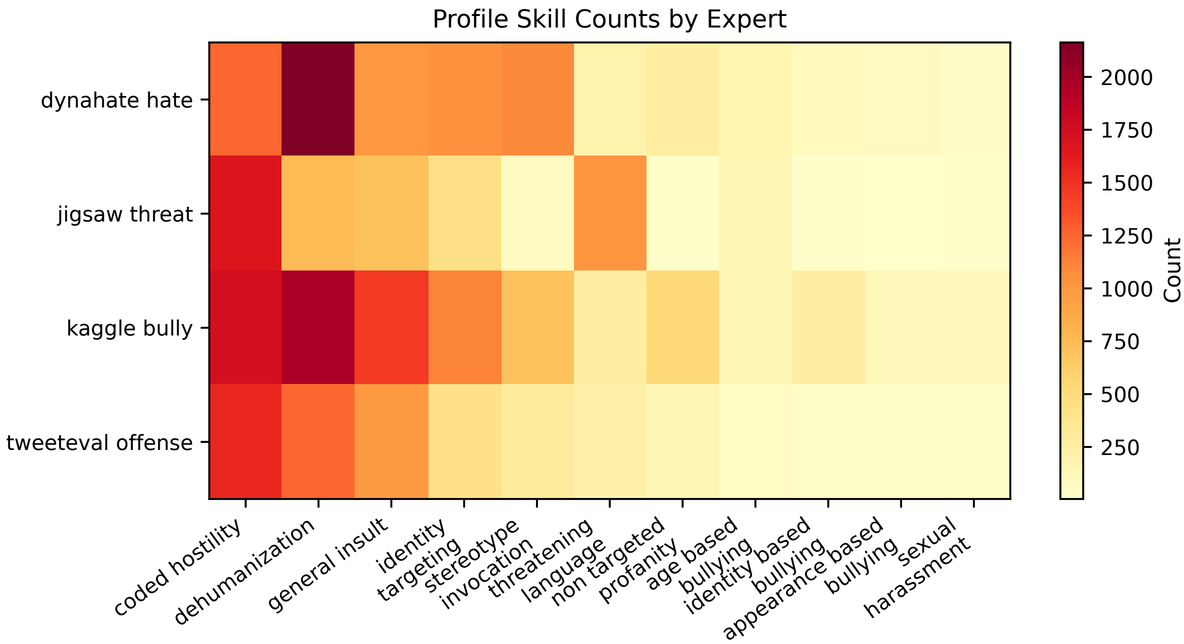


Figure 5.3: Expert Profile Heatmap

The statistics in the expert profile show that the correlations between experts and skills are not trivial but can be interpreted. As shown in Figure 5.6, the hate, threat, and bully experts all exhibit strong correctness counts over several high-frequency skills, while the offense expert remains weaker overall. The hate expert's profile shows that it is particularly strong on dehumanization, coded hostility, stereotype invocation, and identity targeting. The threat expert is strongest on coded hostility and threatening language, while the bully expert performs strongly across a broader range of harmful-language skills, including general insult, identity targeting, and bullying-related subtypes. These tendencies suggest that the profile layer carries significant competence associations between experts and skills.

Simultaneously, the profiles are not completely exclusive. Several common skills, particularly coded hostility, dehumanization, and general insult, are associated with high totals and high correct counts

Table 5.5: Profile Statistics Summary

Expert	Total profile observations	Correct observations	Profile-pool accuracy
Hate Expert	3500	3339	0.9540
Offense Expert	3500	2810	0.8029
Bully Expert	3500	3372	0.9634
Threat Expert	3500	3405	0.9729

across more than one expert rather than belonging to a single one. In most scenarios, one expert might be at their best with a specific ability, yet others will be equally qualified with the same circumstance. This is an indication that the profiles encode conditional competence more explicitly than they encode expert exclusivity.

This difference can be significant in the context of grasping the role of profiles within the entire framework. The profile layer does not directly respond to the question of what skill is unique to this or that expert. Rather it gives the less glamorous, yet equally useful, information of the predictability of a certain expert when a certain symbolic skill is involved. Accordingly, the profiles serve as a layer of descriptive competence and as a provider of soft routing evidence. They thus contribute to the explanation of why routing has significant signal and yet still trying to produce sharp expert separation where several experts are still in a position to be competent in the same common skills.

The conclusion of the profile results is thus cautious. These expert profiles have a meaning, can be interpreted, and are structured adequately to warrant their presence in the routing pipeline. Meanwhile, they also display high-level expertise in dealings with overlapping harmful-language signals, as opposed to a rigidly separated area of expertise. The profile layer within the existing paradigm is thus more prop-

Table 5.6: Validation overall comparison across routing settings.

Setting	Overall Acc	Routing Top-1	Gold Top-1	Gold Top-2	Gold Top-3	Macro F1	Micro F1
Baseline	0.8581	0.6353	0.3519	0.5076	0.7500	0.9097	0.9448
NB Top-1	0.6816	0.6816	0.4461	0.4461	0.4461	0.9446	0.9606
NB Top-2	0.8031	0.6816	0.4461	0.7130	0.7130	0.9300	0.9390
NB Top-3	0.8584	0.6816	0.4461	0.7130	0.8668	0.9253	0.9280

Table 5.7: Validation dataset accuracy by routing setting.

Dataset	Baseline	NB Top-1	NB Top-2	NB Top-3
DynaHate	0.9412	0.7375	0.8333	0.9096
Jigsaw Threat	0.9619	0.8268	0.8725	0.8889
SOSNet Cyberbullying	0.9401	0.5664	0.7800	0.8410
TweetEval Offensive	0.5893	0.5959	0.7266	0.7941

erly viewed as an informative and interpretable routing mechanism than as a generator of robust expert separability.

5.5 End-to-End Routed System Performance

This subsection checks the routed pipeline at the post level against the validation and test pool held out. The fact that the framework uses many experts on a post-by-post basis suggests that end-to-end performance is the result of multi-expert prediction as opposed to the output of an individual classifier. Post-level overall accuracy and per-dataset accuracy and expert level summary metrics are thus used to report performance.

With the ultimate Bernoulli routing setup of top-2 selection, the system has a general validation precision of 0.8031 and a test precision of 0.8135. The associated macro F1 scores are 0.9300 in case of

Table 5.8: In-domain expert accuracy and F1 by routing setting.

Expert	Setting	Accuracy	F1
DynaHate Expert	Baseline	0.9412	0.9463
DynaHate Expert	NB Top-1	0.9482	0.9644
DynaHate Expert	NB Top-2	0.9409	0.9581
DynaHate Expert	NB Top-3	0.9400	0.9477
Offense Expert	Baseline	0.8530	0.7663
Offense Expert	NB Top-1	0.7725	0.8491
Offense Expert	NB Top-2	0.8218	0.8083
Offense Expert	NB Top-3	0.8040	0.8086
Cyberbullying Expert	Baseline	0.9401	0.9634
Cyberbullying Expert	NB Top-1	0.9829	0.9912
Cyberbullying Expert	NB Top-2	0.9719	0.9855
Cyberbullying Expert	NB Top-3	0.9533	0.9754
Threat Expert	Baseline	0.9619	0.9629
Threat Expert	NB Top-1	0.9762	0.9738
Threat Expert	NB Top-2	0.9694	0.9682
Threat Expert	NB Top-3	0.9700	0.9694

validation and 0.9276 in case of test with the corresponding micro F1 scores being 0.9390 and 0.9371 respectively. These figures show that the routed pipeline is quite robust between the two end points at the chosen top-k, although its performance is not evenly distributed between tasks.

This imbalance is shown by the per-dataset results. Bernoulli top-2 system on the validation pool achieves the accuracies of 0.9096 with DynaHate, 0.8889 with Jigsaw Threat and 0.7941 with TweetEval-Offensive at top-2, but the top-2 yields the overall accuracy of 0.8031 with notably high increases in offensive-language coverage when compared to the previous baseline. In the test pool, the chosen top-2 setting has achieved a per-dataset accuracy of 0.8000 in DynaHate, 0.9229 in Jigsaw Threat, 0.7937 in SOSNet Cyberbullying, and 0.7375 in TweetEval-Offensive. The Bernoulli router significantly enhances

Table 5.9: Test overall comparison across routing settings.

Setting	Overall Acc	Routing Top-1	Gold Top-1	Gold Top-2	Gold Top-3	Macro F1	Micro F1
Baseline	0.8500	0.6583	0.3531	0.5042	0.7500	0.9086	0.9333
NB Top-1	0.5708	0.5708	0.3297	0.3297	0.3297	0.9065	0.8970
NB Top-2	0.8135	0.6698	0.4130	0.7177	0.7177	0.9276	0.9371
NB Top-3	0.8682	0.6698	0.4130	0.7177	0.8740	0.9224	0.9266

Table 5.10: Test dataset accuracy by routing setting.

Dataset	Baseline	NB Top-1	NB Top-2	NB Top-3
DynaHate	0.9042	0.8000	0.8000	0.8792
Jigsaw Threat	0.9625	0.5167	0.9229	0.9250
SOSNet Cyberbullying	0.9313	0.2437	0.7937	0.8562
TweetEval Offensive	0.6021	0.7229	0.7375	0.8125

the offensive-language task, relative to the previous skill-log-odds baseline, although not all individual datasets enhance.

These findings are to be interpreted along with the evaluation rule applied in this study. A post is correct, in case a routed expert gives the gold normalized label, and negative labels are also correct, in case the positive label of the prediction does not appear in all the routing predictions. The stated end-to-end accuracy is thus designingly liberal and incentivizes wider expert coverage. This is why the overall accuracy should not be used as the indicator of selective routing quality. As an alternative, it is to be read together with the routing-oriented analyses in the next subsection.

Table 5.11: In-domain expert accuracy and F1 by routing setting on test.

Expert	Setting	Accuracy	F1
DynaHate Expert	Baseline	0.9042	0.9109
DynaHate Expert	NB Top-1	0.8918	0.9239
DynaHate Expert	NB Top-2	0.8918	0.9239
DynaHate Expert	NB Top-3	0.9144	0.9221
Offense Expert	Baseline	0.8810	0.8052
Offense Expert	NB Top-1	0.8786	0.8581
Offense Expert	NB Top-2	0.8517	0.8324
Offense Expert	NB Top-3	0.8273	0.8251
Cyberbullying Expert	Baseline	0.9313	0.9573
Cyberbullying Expert	NB Top-1	0.8947	0.9375
Cyberbullying Expert	NB Top-2	0.9767	0.9878
Cyberbullying Expert	NB Top-3	0.9551	0.9758
Threat Expert	Baseline	0.9625	0.9609
Threat Expert	NB Top-1	–	–
Threat Expert	NB Top-2	0.9665	0.9663
Threat Expert	NB Top-3	0.9667	0.9665

5.6 Routing Behavior and Expert Selection

Although end-to-end accuracy is a measure of the joint performance of routing and expert prediction, it alone does not imply that routing is choosing experts sensibly. Due to that, routing quality has been considered independently with routing top-1 accuracy and gold-expert recall@k. Routing top-1 is a measurement that the number-1 ranked expert is sufficient on its own under the post-level evaluation rule, and gold-expert recall@k is a measurement that the dataset-matched expert is observed in the top-k routed experts.

The Bernoulli router has significant improvements on expert ranking compared to the previous skill-log-odds baseline. At the validation pool, routing top-1 becomes 0.6816 as compared to 0.6353 and gold-

Table 5.12: NB Top-k Calibration candidate summary.

Run	Gold Recall@k	Routing Top-1	Overall Acc	Macro F1	Micro F1	Eligible
Top-1	0.4461	0.6816	0.6816	0.9446	0.9606	no
Top-2	0.7130	0.6816	0.8031	0.9300	0.9390	yes
Top-3	0.8668	0.6816	0.8584	0.9253	0.9280	yes

expert recall becomes 0.4461 as compared to 0.3519 at top-1 and top-2 respectively. Routing top-1 reaches higher from 0.6583 to 0.6698 and gold-expert recall rises at 0.4130 at top-1 and higher 0.7177 at top-2. These improvements show that the revised router is more effective in putting the right expert at the top of the ranking, despite the fact that the post-level accuracy at the end of the line is not improving consistently across data sets.

The tradeoff between selectivity and coverage can also be observed in the top -k comparison. Top-1 setting is the most challenging environment and offers the hardest test of routing accuracy. Top-3 execution provides the best gold-expert coverage, and the best permissive end-to-end accuracy, but is more of a high-coverage routing than a highly selective expert gate. Top-2 thus becomes the most justifiable trading off: the minimal setting which satisfies the validation gold-expert recall criterion and still maintains the routing top-1 performance hence the choice of this pipeline as the final operating point.

Based on per-dataset routing analysis, this improvement is not even. The largest improvements are achieved on offensive-language routing, where the baseline router mis-ranked the offense expert less often whereas Bernoulli router ranked the expert more often in the top-rankings. Nevertheless, the process of routing is still flawed, especially in instances where general skills that can be coded, dehumanization, and general insult control the symbolic expression. When this happens, there are several experts who are still

Table 5.13: Routing behavior summary across baseline and Bernoulli settings. The selected-k column records the explicit execution depth for NB runs; the baseline router uses a variable thresholded expert set rather than a fixed top-k rule.

Setting	Split	Routing Top-1	Gold@1	Gold@2	Gold@3	Selected k
Baseline	Validation	0.6353	0.3519	0.5076	0.7500	–
NB Top-1	Validation	0.6816	0.4461	0.4461	0.4461	1
NB Top-2	Validation	0.6816	0.4461	0.7130	0.7130	2
NB Top-3	Validation	0.6816	0.4461	0.7130	0.8668	3
Baseline	Test	0.6583	0.3531	0.5042	0.7500	–
NB Top-1	Test	0.5708	0.3297	0.3297	0.3297	1
NB Top-2	Test	0.6698	0.4130	0.7177	0.7177	2
NB Top-3	Test	0.6698	0.4130	0.7177	0.8740	3

viable matches, and routing is more like ranked selection of related experts than like expert partitioning which is sharply sparse.

This limitation is further explained by empty-skill cases. A significant proportion of the posts have no retained symbolic skills, limiting the quantity of immediate positive evidence to the router. In the Bernoulli model, they are still represented by priors along with absence evidence in the skill vocabulary, but are nevertheless diagnostically significant, in the sense of diluting the effect of explicit symbolic evidence. The combination of these results is that routing within the existing framework is not only significant, but also nontrivial, but also only partially selective. It has its major strength in the fact that it generates a helpful expert ranking when the evidence it is based on is overlapping symbolically and not in the fact that it enforces a clean mixture-of-experts partition.

5.7 Interpretability of Symbolic Routing

One of the central contributions of the proposed framework is interpretability by design. Rather than linking raw text directly to expert selection through a fully latent mechanism, the system exposes an intermediate symbolic layer that can be inspected at each stage of decision making. Concretely, each prediction follows a traceable chain: post text $\rightarrow$ inferred skills $\rightarrow$ ranked/routed experts $\rightarrow$ expert predictions. Because routing is explicit in this architecture, each step can be analyzed independently. This transparency is especially valuable for harmful-language moderation, where category boundaries are often ambiguous and overlap across tasks is common.

This symbolic-to-expert chain is supported not only qualitatively but also quantitatively. First, inferred skill distributions across datasets (Section 5.3) show semantically meaningful structures rather than random allocations. Second, expert profiles (Section 5.4) provide interpretable summaries of skill-conditioned competence, clarifying why some experts are favored under specific symbolic conditions. Third, routing metrics such as top-1 accuracy and gold-expert recall (Section 5.6) indicate that ranking is not arbitrary: higher-ranked experts are more likely to match task-relevant experts. Finally, the Bernoulli routing mechanism improves ranking quality over the skill-log-odds baseline, showing that symbolic evidence is not only readable but also operationally useful for expert selection.

This claim is illustrated in Table 5.14, which trace representative posts through the full routing pipeline. In successful cases, inferred skills align closely with observable harmful-language content, and routed experts correspond to plausible task-specific interpretations. For example, posts with identity-targeting or dehumanizing language tend to activate experts aligned with hate detection, while posts containing explicit threats tend to activate the threat expert. The comparison cases also make configuration effects

Table 5.14: Representative qualitative cases used to illustrate successful routing, routing drift, subtype loss, and empty-skill failure.

Case Type	Dataset	Gold Label	Inferred Skills
Successful routing	DynaHate	hate	dehumanization, stereotype invocatic
Successful offensive case	TweetEval Offensive	offensive	non-targeted profanity
Routing-drift failure	DynaHate	hate	age-based bullying
Subtype absorption failure	SOSNet Cyberbullying	bully (gender subtype)	dehumanization, coded hostility
Empty-skill failure	DynaHate	hate	none retained

visible: baseline and NB-top2 routing can produce different expert rankings for the same post, allowing direct inspection of how routing decisions change across strategies.

Importantly, interpretability is preserved even when routing is imperfect. When selected experts do not fully align with the gold task, the symbolic layer still provides diagnostic value. Ambiguous posts often trigger overlapping high-frequency skills such as coded hostility and general insult (Section 5.3), and these skills are associated with multiple experts in the profile layer (Section 5.4). In such cases, misrouting can be interpreted as an effect of overlapping symbolic evidence rather than opaque model behavior. This supports targeted error analysis by distinguishing potential causes such as skill-inference ambiguity, profile overlap, or thresholding effects.

Overall, interpretability in this framework is not a superficial add-on but a direct result of the system design. By exposing a structured pathway from text to skills, from skills to expert ranking, and from ranking to final predictions, the framework enables practical auditing of both successful and failure cases. The routing system provides meaningful expert ranking grounded in symbolic evidence, while the symbolic layer keeps these decisions interpretable even when expert selection is not perfect.

5.8 Limitations and Error Analysis

This subsection will summarize major limitations of the current framework and provide an error analysis of representative failure modes at both expert and routing levels.

5.9 Research Question Summary

This subsection will synthesize findings from Sections 5.1–5.7 and provide direct answers to the thesis research questions.

Chapter 6

Conclusion

This is where your Conclusion will go

Appendix A

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Bibliography

Navaud, G. (2011). *Persona : le théâtre comme métaphore théorique de Socrate à Shakespeare.* Droz. http://proxy-remote.galib.uga.edu/login?url=http://search.ebscohost.com/login.aspx?direct=true&db=cat06564a&AN=uga.9939271873902959&site=eds-live

Whitmire, C., Rokad, B., & Crumley, C. (2019). Origami inspired solar panel design. *CoRR*, *abs/1905.06012*. http://arxiv.org/abs/1905.06012